A SILENT ARMADA

Unmanned Surface Vessels (USVs) and Unmanned Underwater Vehicles (UUVs) can transform naval operations. India is developing some of its own, while other countries are at a more advanced stage

BRAINBOX SMART SHIP OPERATIONS
Developed by Tardid Technology, the USV is 8 m long, with endurance of 12 hrs and speed of 12 nmph. To be used for surveillance missions, but can be weaponised too. Also has stealth capability, vast array of sensors, high bandwidth satellite data link networks



ARMED AUTONOMOUS UNCREWED BOAT
Developed by Sagar Defence Engineering Ltd, this lethal USV has swarming capability, AI-based collision avoidance, and smart vision capability suites



REMOTELY OPERATED UNDERWATER VEHICLE, AUTONOMOUS UNDERWATER VEHICLE
These UUVs are being developed by L&T Defence, MDL and DRDO. They have swarm technology, endurance of 10-20 hours. Per global standards, lightweight AUVs are compatible with torpedo tubes on ships/ submarines, endurance of two days; heavy, high-endurance AUVs can remain submerged for as many as 15 days

The Indian navy came up with a global tender last year to acquire 40 NUAS for warships over 100 metres in length. The unmanned aerial systems would be used for surveillance, including signals intelligence (SIGINT) and maritime domain awareness around a naval task force. The tender also mentioned secondary responsibilities like helping with search and rescue, anti-piracy and anti-terrorist operations. The process of acquiring 10 ship-borne NUAS worth around Rs 1,300 crore is on the fast track mode. At present, the Indian navy operates the Israeli-made Heron and Searcher Mk-II drones for reconnaissance. However, ship-borne NUAS are a cut above—they are equipped with a wider range of payloads, including high-resolution synthetic aperture radars, IR systems, electronic intelligence (ELINT), SIGINT and situational awareness systems, as well as anti-surface and anti-ship munitions. The US Navy's MQ-4C Triton and MQ-8B Fire Scout and China's Guizhou Xianglong (Soar Dragon) are the most advanced aerial systems. The navy is also likely to shop for foreign-made UUVs.

In 2021, former navy chief Admiral Karambir Singh had mentioned 'underwater domain awareness' as a critical area, mentioning the 'Chinese' autonomous underwater gliders found lurking in Indonesian waters in 2020. Soon after this, in October 2021, Vice Admiral Ashok Kumar, then the vice chief of the navy and now India's first national maritime security coordinator (NMSC), unveiled a detailed roadmap of the navy's requirements for unmanned platforms and shared it with the domestic industry.

Commodore Anil Jai Singh, a former submariner who is now the vice president of the Indian Maritime Foundation, says that unmanned platforms will play a significant role as force multipliers. "USVs and UUVs that are being developed can complement the existing submarine capability. Their modular configuration gives them the flexibility to carry different payloads, thus making them ideal for a variety of roles and missions," he says, adding that they will be an integral part of a 'networked' battlespace in littoral as well as blue waters and are an asset for enhancing both maritime, including underwater, domain awareness.

According to a top official at the naval headquarters in New Delhi, "The current global environment is witnessing an explosive growth of autonomous systems, which are poised to revolutionise warfare. Unmanned and autonomous systems will create fundamental shifts in the way warfare is conducted."

THE CHINESE THREAT

India's growing emphasis on unmanned capability for its navy has much to do with the clear and present danger from the Chinese navy's expansionist moves. In a bid to counter their growing presence in the Indian Ocean, the Indian Navy has been taking steps to improve its surveillance. USVs, UUVs and ship-borne aerial systems can be a game-changer in this.

Recently, navy chief Admiral R. Harikumar said that at any point in time, there are three-six Chinese warships in the Indian Ocean Region (IOR), from the Gulf of Oman to its eastern fringes. Chinese research vessels and fishing vessels are also ubiquitous. "So, we refine our plans of actions and this feeds into our capability development," he said. China gets unfettered access to the IOR through Pakistan's Gwadar port, which it is developing under the China Pakistan Economic Corridor project, and its possession of Sri Lanka's Hambantota port affords it a presence in India's perceived backyard. It also has a military base in the Indian Ocean in Djibouti. China's footprint in naval unmanned technology is growing. In January 2023, it launched Zhu Hai Yun—the world's first unmanned drone carrier, which can both be controlled